Pre-analysis plan for the #Manydaughters Project

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1 Introduction

This document presents a pre-analysis plan (PAP) to test the four hypotheses outlined in the #Manydaughters Project instructions. For each hypothesis, I describe what variables will be analyzed, what inclusion criteria will be used, what analysis will be performed, and how the results will be interpreted.

2 A note on the general estimation strategy

The general strategy I will use to test for the causal effect of having daughters on a given outcome will be to examine how the outcome differs between respondents who have an additional daughter and respondents who have an additional son. One might define the variable "having an additional daughter" in various ways; for example, as the number of daughters conditional on the number of children or as an indicator that the respondent has at least one daughter conditional on the number of children. In either case, conditioning on the total number of children is important because it prevents confounding the effect of having daughters and the effect of having children per se (as noted by Washington, 2008; Glynn and Sen, 2015; and Ronchi and Smith, 2024).

The main identifying assumption in this strategy is that the gender of the child is random conditional on the total number of children. This assumption would be invalidated if couples have a preference for a particular gender of the child—for example, a son preference or a preference for gender balance—and can act on this preference by selecting the gender of the child, for example by adopting a child of the desired gender or by using a gender-biased fertility stopping rule or a technology for fetal sex selection. To reduce selection concerns, I will restrict the sample to biological children as opposed to adopted children. To further reduce concerns about selection due to a preference for gender balance, I will define "having an additional daughter" as having a firstborn child who is a daughter conditional on having only one child. That is, I will examine how the outcome of interest differs between respondents who have a firstborn child who is a daughter and respondents who have a firstborn child who is a son, conditional on having only one child. Following Bhalotra, Clarke, and Nazarova (2025), it is important to test for the validity of the assumption that the gender of the child is random by checking for balance in observable characteristics of the respondent (e.g., age, marital status, educational attainment, and household income)

between respondents who have a daughter and respondents who have a son in this sample. However, since the #Manydaughters Project does not allow one to submit analysis other than one single regression-based test per hypothesis, I do not submit the balance check.

3 Variables

I defer discussing the outcome variables till I present each hypothesis. Unless otherwise noted, the independent variables mentioned in Section 5 are defined as follows.

- *Daughter indicator*: indicator that equals 1 if the child is female (and 0 if the child is male)

- *Gender of the respondent*: indicator that equals 1 if the respondent is female (and 0 if the respondent is male)

- *Household income*: household income of the respondent based on SOEP item hl/hlc0005_h: [de] Monatliches HH-Netto-Einkommen (Euro) [harmonisiert]

- *Age of the respondent*: age of the respondent at the time of taking the survey

- *Educational attainment*: highest degree attained by the respondent based on SOEP item pgen/pgisced: Highest degree/diploma attained, ISCED-1997

4 Inclusion criteria

As mentioned in Section 2, the analysis will restrict the sample to respondents who have a firstborn biological child conditional on having only one child. Only children classified as male or female will be included in the sample. Data from all available years in the period 2000-2020 will be included.

5 Exact analysis

5.1 Main analysis

The analysis will be cross-sectional; namely, it will compare respondents at any given time point in the data. Unless otherwise noted, the analysis will regress using OLS the outcome variable on an indicator that the child is female, controlling for the gender of the respondent, the age of the respondent, the age of the respondent squared, household income of the respondent, educational attainment of the respondent, and survey year fixed effects. I will estimate heteroskedasticity robust standard errors and weight observations using sampling weights provided by SOEP to create country population parameter estimates.

6 Interpretation of the results

I will regard a result as statistically significant if the p-value for the coefficient of interest in a two-sided test is less than 0.05.

7 Hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

7.1 Outcome variable

Gender equality refers to the relative position of men and women in society in terms of education, health, politics, and the labor market (Duflo, 2012; Blau and Winkler, 2022). Attitudes toward gender equality are often measured in surveys by asking respondents for their views of the appropriate place of women in society on those dimensions. For example, the World Values Survey (Haerpfer, 2020) and the UNDP Gender Social Norms Index (UNDP, 2020) ask respondents to rate their agreement with statements such as: "*Men make better political leaders than women do*", "*Women have the same rights as men*", "*University is more important for a man than for a woman*", "*Men should have more right to a job than women*", "*Men make better business executives than women do*", "*Family suffer if the mother works*", "*A man's job is to earn money; a woman's job is to look after the home and family*".

The only item I have been able to find in the SOEP documentation that resembles the statements above is:

> GIP_W39_V2/AC39161: responsibility gender equality (raw answers 0-10)
>
> German: Sollte der Staat Ihrer Meinung nach dafür verantwortlich sein, die Chancengleichheit von Männern und Frauen im Berufsleben sicherzustellen?
>
> English: In your opinion, the state should be responsible for ensuring equal opportunities for men and women in professional life?

I will use the raw values to the item *responsibility gender equality* as the outcome measure of attitudes toward gender equality. If after receiving access to the SOEP dataset I find other items in SOEP that resemble the survey statements above that measure attitudes toward gender equality, I will use as outcome variable either the raw values of the SOEP item that most closely resembles the statements above, or an overall index of gender-equality attitudes that is the average response to all such SOEP items.

8 Hypothesis 2

H2. *Having daughters leads to more left-leaning political preferences.*

8.1 Outcome variable

I will measure left-leaning political preferences with an indicator for not having a preference for a right-wing party based on the SOEP item:

pl/plh0012_h: [de] Parteineigung [harmonisiert]

The indicator will equal 0 if the party is right-wing, and 1 otherwise.

9 Hypothesis 3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*

9.1 Outcome variable

It is difficult to test gender equality in the household with a single regression test, because gender equality involves various dimensions including division of market work and home production in the household and health of the mother. Since the #Manydaughters project asks one to submit one single regression-based test per hypothesis, I choose as outcome the mother's employment status. In particular, whether the mother is employed full time.

- *Mother employed full time:* Indicator that equals 1 if the mother is employed full time and 0 otherwise. Greater likelihood that the mother is employed full time indicates greater gender equality. Employment status will be determined based on the SOEP item pl/plb0022_h: [de] Erwerbsstatus [harmonisiert].

10 Hypothesis 4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

10.1 Outcome variable

I will use the same outcome variable as in Hypothesis 1.

10.2 Exact analysis

The analysis to test Hypothesis 4 will be identical to that outlined in Section 2 except that it will add a term that interacts the daughter indicator and the gender of the parent. More specifically, the analysis will regress using OLS the outcome variable on an indicator that the child is female, the gender of the respondent, and the interaction of the two, controlling for the age of the respondent, the age of the respondent squared, household income of the respondent, educational attainment of the respondent, and survey year fixed effects. I will estimate heteroskedasticity robust standard errors and weight observations using sampling weights provided by SOEP to create country population parameter estimates.

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